

Step-by-Step SVM Derivation (Beginner Friendly)

SVM — Linear Example: Full step-by-step with plots and arithmetic

Summary of given data

Positive class (label +1):

$$(3, 1), (3, -1), (6, 1), (6, -1).$$

Negative class (label -1):

$$(1, 0), (0, 1), (0, -1), (-1, 0).$$

By inspection the sample is linearly separable; the worked example chooses three support vectors:

$$s_1 = (1, 0) \ (y_1 = -1), \quad s_2 = (3, 1) \ (y_2 = +1), \quad s_3 = (3, -1) \ (y_3 = +1).$$

We will:

- plot the points and highlight the support vectors,
- augment vectors to include bias and compute all dot-products,
- derive and solve the 3×3 linear system (every arithmetic step),
- compute $\tilde{w}$, extract w and b , and plot the decision line,
- verify the margin equalities for support vectors.

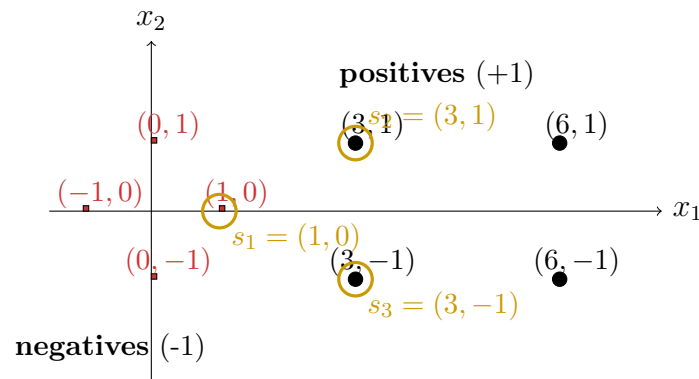


Figure 1: Input space: blue = positive class, red = negative class; support vectors highlighted in orange.

1 Augment vectors to include bias

To absorb bias b into the weight vector we augment each vector with a 1:

$$\tilde{s}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \tilde{s}_2 = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}, \quad \tilde{s}_3 = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}.$$

We will use augmented weight $\tilde{w} = (w_1, w_2, b)^\top$ and write constraints using $\tilde{w}^\top \tilde{s}$.

2 Derivation of the 3×3 system (why these equations)

For a support vector s_k (on the margin) the KKT equality gives:

$$y_k(\tilde{w}^\top \tilde{s}_k) = 1.$$

Using $\tilde{w} = \sum_i \alpha_i \tilde{s}_i$ (optimality condition), substitute:

$$y_k \left(\left(\sum_i \alpha_i \tilde{s}_i \right)^\top \tilde{s}_k \right) = 1 \quad \implies \quad \sum_i \alpha_i (\tilde{s}_i \cdot \tilde{s}_k) = y_k,$$

where we defined α_i as the signed combination used in the worked example. This is the linear system solved for α_i using only the support vectors (i.e. $i, k \in \{1, 2, 3\}$).

3 Compute all dot products (show arithmetic)

Compute each augmented dot product $\tilde{s}_i \cdot \tilde{s}_j$ component-wise (sum of component products).

$$\tilde{s}_1 \cdot \tilde{s}_1 = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = 1 + 0 + 1 = 2,$$

$$\tilde{s}_2 \cdot \tilde{s}_1 = 3 \cdot 1 + 1 \cdot 0 + 1 \cdot 1 = 3 + 0 + 1 = 4,$$

$$\tilde{s}_3 \cdot \tilde{s}_1 = 3 \cdot 1 + (-1) \cdot 0 + 1 \cdot 1 = 3 + 0 + 1 = 4,$$

$$\tilde{s}_2 \cdot \tilde{s}_2 = 3 \cdot 3 + 1 \cdot 1 + 1 \cdot 1 = 9 + 1 + 1 = 11,$$

$$\tilde{s}_2 \cdot \tilde{s}_3 = 3 \cdot 3 + 1 \cdot (-1) + 1 \cdot 1 = 9 - 1 + 1 = 9,$$

$$\tilde{s}_3 \cdot \tilde{s}_3 = 3 \cdot 3 + (-1) \cdot (-1) + 1 \cdot 1 = 9 + 1 + 1 = 11.$$

4 Form the 3×3 linear system

Plug dot-products and labels $y_1 = -1$, $y_2 = +1$, $y_3 = +1$ into $\sum_i \alpha_i (\tilde{s}_i \cdot \tilde{s}_k) = y_k$ for $k = 1, 2, 3$:

$$\begin{aligned} 2\alpha_1 + 4\alpha_2 + 4\alpha_3 &= -1, \\ 4\alpha_1 + 11\alpha_2 + 9\alpha_3 &= +1, \\ 4\alpha_1 + 9\alpha_2 + 11\alpha_3 &= +1. \end{aligned}$$

We will now solve this system exactly, showing every arithmetic operation.

5 Solve the system by Gaussian elimination (detailed arithmetic)

Write the augmented matrix form:

$$\left[\begin{array}{ccc|c} 2 & 4 & 4 & -1 \\ 4 & 11 & 9 & +1 \\ 4 & 9 & 11 & +1 \end{array} \right]$$

Step 1: Eliminate α_1 from row 2 and row 3

Pivot is row1 first entry 2. We eliminate the 4 in rows 2 and 3 by subtracting $2 \times$ row1.

Row2 $\leftarrow$ Row2 $-2 \times$ Row1:

$$\begin{array}{lcl} & 4 - 2 \cdot 2 & = 0, \\ \text{Row2 :} & 11 - 2 \cdot 4 & = 11 - 8 = 3, \\ & 9 - 2 \cdot 4 & = 9 - 8 = 1, \\ & 1 - 2(-1) & = 1 + 2 = 3. \end{array}$$

So new Row2 is $[0 \quad 3 \quad 1 \mid 3]$.

Row3 $\leftarrow$ Row3 $-2 \times$ Row1:

$$\begin{array}{lcl} & 4 - 2 \cdot 2 & = 0, \\ \text{Row3 :} & 9 - 2 \cdot 4 & = 9 - 8 = 1, \\ & 11 - 2 \cdot 4 & = 11 - 8 = 3, \\ & 1 - 2(-1) & = 1 + 2 = 3. \end{array}$$

So new Row3 is $[0 \quad 1 \quad 3 \mid 3]$.

The matrix is now:

$$\left[\begin{array}{ccc|c} 2 & 4 & 4 & -1 \\ 0 & 3 & 1 & 3 \\ 0 & 1 & 3 & 3 \end{array} \right]$$

Step 2: Eliminate α_2 from row 3

We want to make the (3,2) entry zero. Use Row2 (pivot is 3 at (2,2)). Multiply Row2 by $\frac{1}{3}$ and subtract from Row3, or equivalently do $\text{Row3} \leftarrow \text{Row3} - \frac{1}{3} \cdot \text{Row2}$.

Compute:

$$\begin{array}{lcl} \text{New (3,2)} & = 1 - \frac{1}{3} \cdot 3 & = 1 - 1 = 0, \\ \text{New (3,3)} & = 3 - \frac{1}{3} \cdot 1 & = 3 - \frac{1}{3} = \frac{9}{3} - \frac{1}{3} = \frac{8}{3}, \\ \text{New RHS} & = 3 - \frac{1}{3} \cdot 3 & = 3 - 1 = 2. \end{array}$$

So Row3 becomes $[0 \quad 0 \quad \frac{8}{3} \mid 2]$.

Now triangular matrix:

$$\left[\begin{array}{ccc|c} 2 & 4 & 4 & -1 \\ 0 & 3 & 1 & 3 \\ 0 & 0 & \frac{8}{3} & 2 \end{array} \right]$$

Step 3: Solve for α_3

From Row3:

$$\frac{8}{3}\alpha_3 = 2 \implies \alpha_3 = \frac{2}{8/3} = 2 \cdot \frac{3}{8} = \frac{6}{8} = \frac{3}{4} = 0.75.$$

Step 4: Back-substitute to find α_2

Row2: $3\alpha_2 + 1 \cdot \alpha_3 = 3$.

Plug $\alpha_3 = 0.75$:

$$3\alpha_2 + 0.75 = 3 \implies 3\alpha_2 = 3 - 0.75 = 2.25.$$

So

$$\alpha_2 = \frac{2.25}{3} = 0.75.$$

Step 5: Back-substitute to find α_1

Row1: $2\alpha_1 + 4\alpha_2 + 4\alpha_3 = -1$.

Plug $\alpha_2 = \alpha_3 = 0.75$ (note $4 \times 0.75 = 3$):

$$2\alpha_1 + 3 + 3 = -1 \implies 2\alpha_1 + 6 = -1.$$

Thus

$$2\alpha_1 = -7 \implies \alpha_1 = -\frac{7}{2} = -3.5.$$

$\alpha_1 = -3.5, \quad \alpha_2 = 0.75, \quad \alpha_3 = 0.75.$

6 Compute augmented weight vector $\tilde{w}$ and extract w, b

We have

$$\tilde{w} = \sum_{i=1}^3 \alpha_i \tilde{s}_i = \alpha_1 \tilde{s}_1 + \alpha_2 \tilde{s}_2 + \alpha_3 \tilde{s}_3.$$

Plugging numbers:

$$\tilde{w} = -3.5 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + 0.75 \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} + 0.75 \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}.$$

Compute each component carefully:

- (1) $-3.5 \cdot 1 + 0.75 \cdot 3 + 0.75 \cdot 3 = -3.5 + 2.25 + 2.25 = 1.0,$
- (2) $-3.5 \cdot 0 + 0.75 \cdot 1 + 0.75 \cdot (-1) = 0 + 0.75 - 0.75 = 0,$
- (3) $-3.5 \cdot 1 + 0.75 \cdot 1 + 0.75 \cdot 1 = -3.5 + 0.75 + 0.75 = -2.0.$

So

$$\tilde{w} = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}.$$

Interpretation: $\tilde{w} = (w_1, w_2, b)^\top$ so

$$w = (1, 0), \quad b = -2.$$

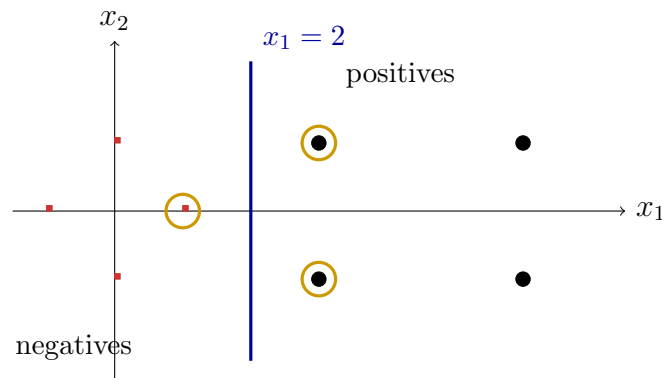
7 Final decision function and plot

Classifier:

$$f(x) = \text{sign}(w^\top x + b) = \text{sign}(x_1 - 2).$$

Decision boundary: $x_1 - 2 = 0 \Rightarrow x_1 = 2$ (vertical line).

Plot the decision boundary together with data (support vectors highlighted):



8 Sanity checks (verify margin equalities)

Compute $w^\top s_k + b$ for each support vector.

$$s_2 = (3, 1) : \quad w^\top s_2 + b = (1, 0) \cdot (3, 1) + (-2) = 3 - 2 = 1.$$

$$s_3 = (3, -1) : \quad w^\top s_3 + b = (1, 0) \cdot (3, -1) - 2 = 3 - 2 = 1.$$

$$s_1 = (1, 0) : \quad w^\top s_1 + b = (1, 0) \cdot (1, 0) - 2 = 1 - 2 = -1.$$

Multiply by labels y_k to check $y_k(w^\top s_k + b) = 1$:

$$y_2(w^\top s_2 + b) = (+1)(1) = 1,$$

$$y_3(w^\top s_3 + b) = (+1)(1) = 1,$$

$$y_1(w^\top s_1 + b) = (-1)(-1) = 1.$$

All margin equalities hold — perfect.

Answer recap:

$\alpha_1 = -3.5, \alpha_2 = 0.75, \alpha_3 = 0.75, \quad w = (1, 0), \quad b = -2, \quad \text{decision line } x_1 = 2.$

- Nonlinear SVM Example — Full step-by-step (with visuals)

Goal

We will follow the nonlinear example where the mapping Φ_1 is nontrivial. We will: (1) show the data in input space, (2) compute (explicitly) Φ_1 on each point when needed, (3) identify support vectors in *feature space*, (4) form the SVM equations, (5) solve them step-by-step, and (6) compute the final classifier $f(x) = \text{sign}(w^\top x + b)$.

Notation: Points in input space are written as (x_1, x_2) . Augmented vectors (to include bias) are written $\tilde{x} = (x_1, x_2, 1)^\top$.

1. Data in input space

Positive (label +1) points:

$$(2, 2), \quad (2, -2), \quad (-2, -2), \quad (-2, 2).$$

Negative (label -1) points:

$$(1, 1), \quad (1, -1), \quad (-1, -1), \quad (-1, 1).$$

Observation: The positives form an outer square (distance from origin $\sqrt{2^2 + 2^2} = \sqrt{8} \approx 2.83 > 2$) surrounding the negatives (inner square, distance $\sqrt{1^2 + 1^2} = \sqrt{2} \approx 1.41 < 2$). No single straight line in input space can separate them.

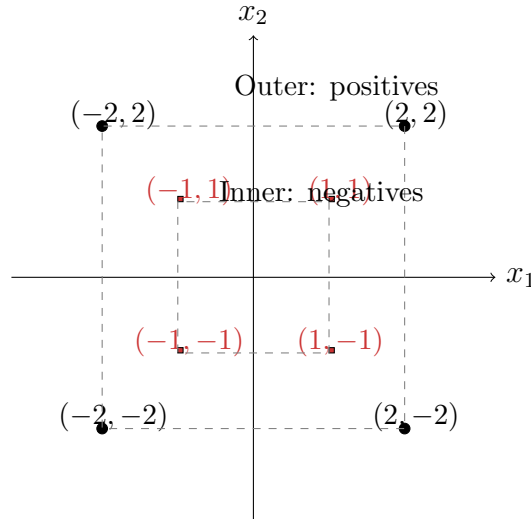


Figure 2: Input space: outer positive square (blue) encloses inner negative square (red).

2. Why we need a mapping Φ

Because the positive and negative classes are concentric, no linear hyperplane in the input space can separate them. We therefore use a nonlinear mapping $\Phi_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ (feature space), chosen so that the classes become linearly separable there.

The mapping given is (piecewise):

$$\Phi_1 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{cases} \begin{pmatrix} 4 - x_2 + |x_1 - x_2| \\ 4 - x_1 + |x_1 - x_2| \end{pmatrix}, & \text{if } \sqrt{x_1^2 + x_2^2} > 2, \\ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, & \text{otherwise.} \end{cases}$$

Intuition: points with radius > 2 (outer square) are “pushed” outward by a nonlinear formula; inner points (radius ≤ 2) remain unchanged.

3. Apply Φ_1 to our points (explicit arithmetic)

We compute Φ_1 for each of the 8 points to be explicit.

Inner (negative) points — radius ≤ 2 so identity

For $(1, 1)$: $\sqrt{1^2 + 1^2} = \sqrt{2} \approx 1.414 < 2 \Rightarrow \Phi_1(1, 1) = (1, 1)$. Similarly,

$$\Phi_1(1, -1) = (1, -1), \quad \Phi_1(-1, -1) = (-1, -1), \quad \Phi_1(-1, 1) = (-1, 1).$$

These stay the same.

Outer (positive) points — radius > 2 so use formula

We compute carefully (do each arithmetic step).

Point $(2, 2)$.

$$\sqrt{2^2 + 2^2} = \sqrt{8} \approx 2.828 > 2 \Rightarrow \text{use first branch.}$$

Compute $|x_1 - x_2| = |2 - 2| = 0$.

$$\Phi_1(2, 2) = \begin{pmatrix} 4 - x_2 + |x_1 - x_2| \\ 4 - x_1 + |x_1 - x_2| \end{pmatrix} = \begin{pmatrix} 4 - 2 + 0 \\ 4 - 2 + 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}.$$

So $(2, 2)$ maps to $(2, 2)$ (it happens to remain the same vector numerically).

Point $(2, -2)$.

$$\sqrt{2^2 + (-2)^2} = \sqrt{8} > 2, \quad |x_1 - x_2| = |2 - (-2)| = 4.$$

Now compute components:

$$\begin{aligned} \text{first component} &= 4 - x_2 + |x_1 - x_2| \\ &= 4 - (-2) + 4 = 4 + 2 + 4 = 10, \end{aligned}$$

$$\begin{aligned} \text{second component} &= 4 - x_1 + |x_1 - x_2| \\ &= 4 - 2 + 4 = 6. \end{aligned}$$

So $\Phi_1(2, -2) = (10, 6)$.

Point $(-2, -2)$.

$$|x_1 - x_2| = |-2 - (-2)| = 0, \quad 4 - x_2 + |x_1 - x_2| = 4 - (-2) + 0 = 6,$$

$$4 - x_1 + |x_1 - x_2| = 4 - (-2) + 0 = 6.$$

So $\Phi_1(-2, -2) = (6, 6)$.

Point $(-2, 2)$.

$$|x_1 - x_2| = |-2 - 2| = 4.$$

Compute:

$$\text{first} = 4 - x_2 + |x_1 - x_2| = 4 - 2 + 4 = 6,$$

$$\text{second} = 4 - x_1 + |x_1 - x_2| = 4 - (-2) + 4 = 10.$$

So $\Phi_1(-2, 2) = (6, 10)$.

Summary of feature-space points:

Positives: $\Phi_1(2, 2) = (2, 2)$, $\Phi_1(2, -2) = (10, 6)$,

$\Phi_1(-2, -2) = (6, 6)$, $\Phi_1(-2, 2) = (6, 10)$.

Negatives: $\Phi_1(1, 1) = (1, 1)$, $\Phi_1(1, -1) = (1, -1)$, $\Phi_1(-1, -1) = (-1, -1)$, $\Phi_1(-1, 1) = (-1, 1)$.

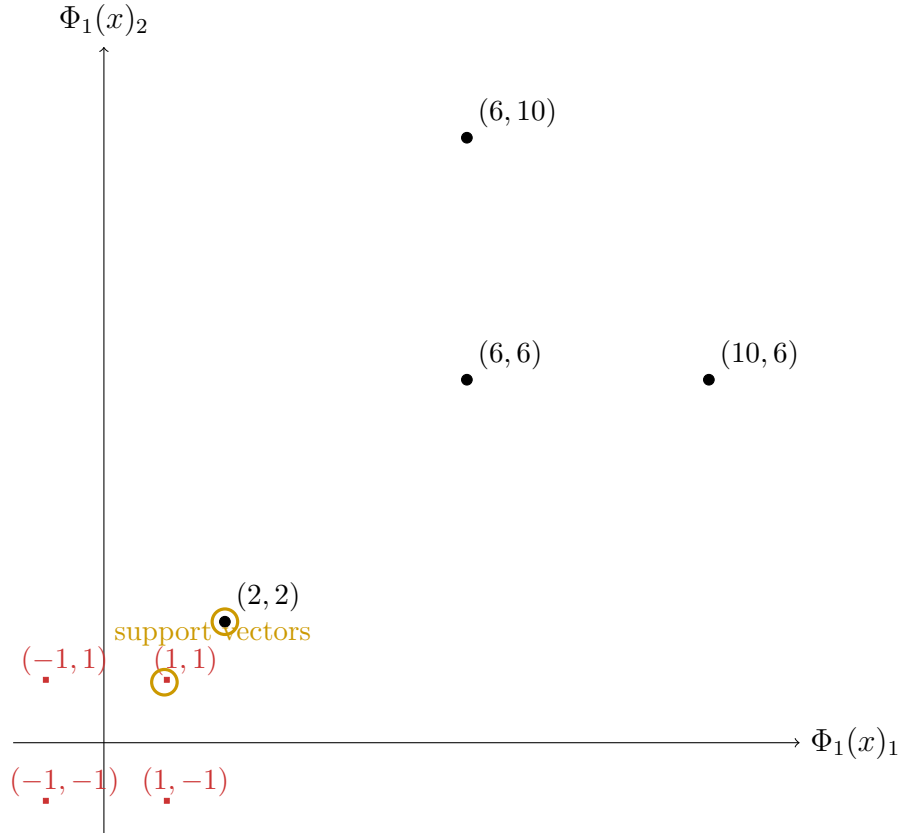


Figure 3: Feature space (after Φ_1). The inner negatives remain near the origin; positives are pushed outward. The points $(1, 1)$ and $(2, 2)$ sit close to the separating boundary and will be chosen as support vectors in this worked example.

4. Choosing support vectors in feature space

Now that the classes are separated in feature space, we look for the points that lie *closest* to the boundary (the margin). Visual inspection shows the negative point $(1, 1)$ (unchanged by Φ_1) and the positive point $(2, 2)$ (one transformed point) are the nearest opposing points. These become our **support vectors**:

$$s_1 = (1, 1) \quad (y_1 = -1), \quad s_2 = (2, 2) \quad (y_2 = +1).$$

We augment them:

$$\tilde{s}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \tilde{s}_2 = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}.$$

5. SVM equalities for support vectors

For a support vector s_k the margin equality holds:

$$y_k(\tilde{w}^\top \tilde{s}_k) = 1.$$

Using $\tilde{w} = \sum_i \alpha_i \tilde{s}_i$ and multiplying both sides by y_k we get:

$$\sum_i \alpha_i (\tilde{s}_i \cdot \tilde{s}_k) = y_k.$$

With two support vectors, this gives a 2×2 linear system in α_1, α_2 :

$$\begin{aligned} \alpha_1(\tilde{s}_1 \cdot \tilde{s}_1) + \alpha_2(\tilde{s}_2 \cdot \tilde{s}_1) &= y_1 = -1, \\ \alpha_1(\tilde{s}_1 \cdot \tilde{s}_2) + \alpha_2(\tilde{s}_2 \cdot \tilde{s}_2) &= y_2 = +1. \end{aligned}$$

6. Compute dot products (explicitly)

Compute each dot product carefully (augment includes bias = 1):

$$\begin{aligned} \tilde{s}_1 \cdot \tilde{s}_1 &= 1^2 + 1^2 + 1^2 = 3. \\ \tilde{s}_2 \cdot \tilde{s}_1 &= (2 \cdot 1) + (2 \cdot 1) + (1 \cdot 1) = 2 + 2 + 1 = 5. \\ \tilde{s}_2 \cdot \tilde{s}_2 &= 2^2 + 2^2 + 1^2 = 4 + 4 + 1 = 9. \end{aligned}$$

So the linear system is:

$$\boxed{\begin{aligned} 3\alpha_1 + 5\alpha_2 &= -1, \\ 5\alpha_1 + 9\alpha_2 &= +1. \end{aligned}}$$

7. Solve the 2×2 system (no shortcuts, step-by-step)

We solve:

$$\begin{bmatrix} 3 & 5 \\ 5 & 9 \end{bmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Method: elimination (multiply subtract). Multiply the first equation by 5 and the second by 3 (so the α_1 coefficient becomes 15 in both):

$$5 \times (3\alpha_1 + 5\alpha_2 = -1) \Rightarrow 15\alpha_1 + 25\alpha_2 = -5,$$

$$3 \times (5\alpha_1 + 9\alpha_2 = +1) \Rightarrow 15\alpha_1 + 27\alpha_2 = +3.$$

Now subtract the first scaled equation from the second scaled equation to eliminate α_1 :

$$(15\alpha_1 + 27\alpha_2) - (15\alpha_1 + 25\alpha_2) = 3 - (-5).$$

Left side simplifies to $2\alpha_2$. Right side simplifies to 8. So:

$$2\alpha_2 = 8 \Rightarrow \alpha_2 = \frac{8}{2} = 4.$$

Substitute $\alpha_2 = 4$ back into the first original equation $3\alpha_1 + 5\alpha_2 = -1$:

$$3\alpha_1 + 5 \cdot 4 = -1 \Rightarrow 3\alpha_1 + 20 = -1.$$

Thus:

$$3\alpha_1 = -1 - 20 = -21 \Rightarrow \alpha_1 = -7.$$

$$\boxed{\alpha_1 = -7, \quad \alpha_2 = 4.}$$

8. Compute the augmented weight vector $\tilde{w}$

$$\tilde{w} = \sum_{i=1}^2 \alpha_i \tilde{s}_i = \alpha_1 \tilde{s}_1 + \alpha_2 \tilde{s}_2.$$

Plug in values:

$$\tilde{w} = -7 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + 4 \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}.$$

Compute component-wise:

$$\text{first component: } -7(1) + 4(2) = -7 + 8 = 1,$$

$$\text{second component: } -7(1) + 4(2) = -7 + 8 = 1,$$

$$\text{third (bias) component: } -7(1) + 4(1) = -7 + 4 = -3.$$

Thus

$$\tilde{w} = \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix}.$$

Interpretation: the first two entries are the weight vector w in input space and the last entry is the bias b :

$$w = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad b = -3.$$

9. Final decision function and boundary

The classifier is:

$$f(x) = \text{sign}(w^\top x + b) = \text{sign}(x_1 + x_2 - 3).$$

The decision boundary (where the score equals 0) is the line:

$$x_1 + x_2 - 3 = 0 \implies x_2 = 3 - x_1.$$

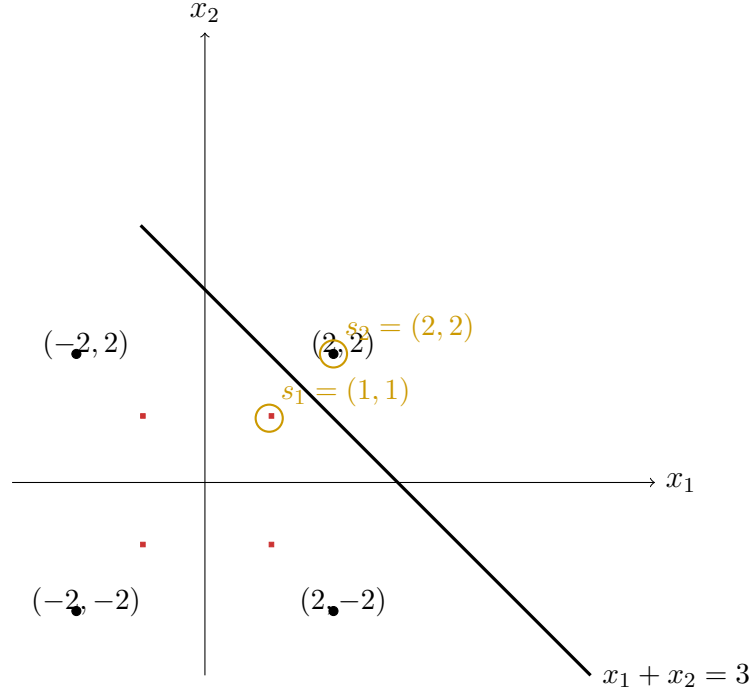


Figure 4: Input space with final decision boundary $x_1 + x_2 - 3 = 0$. The chosen support vectors in input space are highlighted.

10. Quick checks (sanity)

Plug the support vectors into the classifier score $w^\top x + b$:

$$\text{For } s_2 = (2, 2) : w^\top s_2 + b = (1, 1) \cdot (2, 2) + (-3) = 2 + 2 - 3 = 1.$$

$$\text{For } s_1 = (1, 1) : w^\top s_1 + b = 1 + 1 - 3 = -1.$$

These satisfy the margin equalities $y_k(w^\top s_k + b) = 1$ (since $y_2 = +1$ and $y_1 = -1$).

Conclusion: After applying Φ_1 the previously nonlinearly separable data becomes separable in feature space. Choosing the nearest opposite-class points $(1, 1)$ and $(2, 2)$ as support vectors and solving the SVM equalities yields $\alpha_1 = -7, \alpha_2 = 4$ and the final classifier $f(x) = \text{sign}(x_1 + x_2 - 3)$.

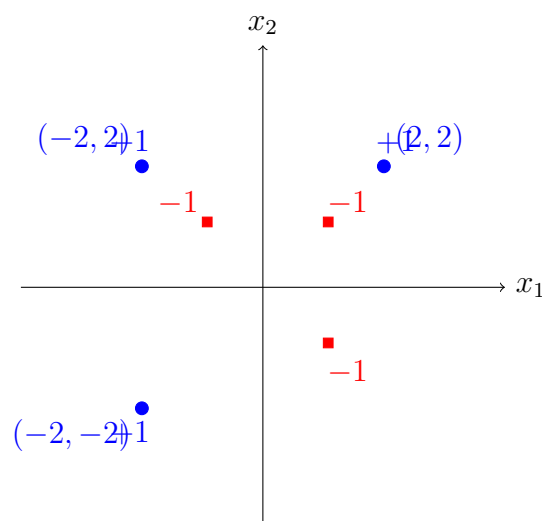
Part (a): Plotting the Dataset and Justification for Hard-Margin SVM

We are given the following labeled dataset:

Label y	Data Point (x_1, x_2)
+1	$(-2, -2)$
+1	$(-2, 2)$
+1	$(2, 2)$
-1	$(1, 1)$
-1	$(1, -1)$
-1	$(-1, 1)$

Plotting the Points in the Input Space

The positive points ($y = +1$) lie towards the outer region of the coordinate plane, while the negative points ($y = -1$) lie closer to the center. The plot below illustrates their locations:



Why the Data is Linearly Separable

The negative class forms a compact group around the origin, while the positive class lies farther away. This means a straight line can be drawn that places all positive points on one side and all negative points on the other. Therefore, the dataset is **linearly separable**, and a **hard-margin SVM** can be used.

Choosing the Separating Hyperplane

To find a separating line, consider the quantity $x_1 + x_2$ for each point:

Point	$x_1 + x_2$	Label
$(-2, -2)$	-4	+1
$(-2, 2)$	0	+1
$(2, 2)$	4	+1
$(1, 1)$	2	-1
$(1, -1)$	0	-1
$(-1, 1)$	0	-1

We observe:

- The negative points have values of $x_1 + x_2$ around 0 to 2.
- The positive points have values either less than 0 or greater than 2.

Thus, the midpoint value between the two groups is:

$$\frac{0 + 2}{2} = 1$$

So a separating line is:

$$\boxed{x_1 + x_2 = 1}$$

Conclusion

Since a straight line (specifically $x_1 + x_2 = 1$) perfectly divides the two classes with no misclassifications, the dataset is **linearly separable**. Therefore, using a **hard-margin SVM** is appropriate for this problem.